MH1201 Linear Algebra II

Tutorial 7 (Week 8 & 9) – Problems & Solutions

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Overview

This tutorial emphasizes:

- matrix representations of linear maps from 1-dimensional spaces and the "single-column" principle;
- upper-triangular structure forced by nested span conditions on images of basis vectors;
- infinite-dimensional phenomena: injective but not surjective operators and absence of eigenvalues;
- rotation operators on $\mathbb{R}^2$ and the (non-)existence of real eigenvalues;
- eigenvalues/eigenspaces of concrete linear maps on $\mathbb{R}^2$ and $\mathbb{R}^{2,2}$;
- diagonalizability criteria via eigenspace dimension counts and explicit eigenbases.

Contents

Question 1 (Matrix of $T_v : \mathbb{F} \to V$)

Problem

Let

(i) V be a finite-dimensional vector space over a field $\mathbb{F}$,

(ii) $v \in V$, and

(iii) β be an ordered basis for V.

Define $T_v \in \mathcal{L}(\mathbb{F}, V)$ as follows:

$$\forall \lambda \in \mathbb{F} : \qquad T_v(\lambda) := \lambda \cdot v.$$

Show that

$$[T_v]_\beta^{(1_\mathbb{F})} = [v]_\beta.$$

Remark: $(1_\mathbb{F})$ is an ordered basis for $\mathbb{F}$.

Solution

Method 1: "One-dimensional domain $\Rightarrow$ one column"

Let $\beta = (b_1, \ldots, b_n)$, where $n = \dim(V)$. Since $\mathbb{F}$ is 1-dimensional over itself with ordered basis $(1_\mathbb{F})$, the matrix $[T_v]_\beta^{(1_\mathbb{F})}$ is an $n \times 1$ matrix whose unique column is the coordinate vector of $T_v(1_\mathbb{F})$ in the basis β:

$$[T_v]_\beta^{(1_\mathbb{F})} = \big[\, [T_v(1_\mathbb{F})]_\beta \,\big].$$

But $T_v(1_\mathbb{F}) = 1_\mathbb{F} \cdot v = v$. Hence

$$[T_v]_\beta^{(1_\mathbb{F})} = [v]_\beta.$$

$$\boxed{[T_v]_\beta^{(1_\mathbb{F})} = [v]_\beta}.$$

Method 2: Use the defining identity $[T(x)] = [T]\,[x]$

For any $\lambda \in \mathbb{F}$, the coordinate of λ in the basis $(1_{\mathbb{F}})$ is

$$[\lambda]_{(1_{\mathbb{F}})} = \begin{bmatrix} \lambda \end{bmatrix}.$$

By the fundamental matrix representation identity,

$$[T_v(\lambda)]_\beta = [T_v]_\beta^{(1_{\mathbb{F}})}\,[\lambda]_{(1_{\mathbb{F}})}.$$

But $T_v(\lambda) = \lambda v$, so by linearity of the coordinate map $[\,\cdot\,]_\beta$,

$$[T_v(\lambda)]_\beta = [\lambda v]_\beta = \lambda[v]_\beta.$$

Thus, for all $\lambda \in \mathbb{F}$,

$$\lambda[v]_\beta = [T_v]_\beta^{(1_{\mathbb{F}})}\begin{bmatrix} \lambda \end{bmatrix}.$$

Taking $\lambda = 1_{\mathbb{F}}$ gives

$$[v]_\beta = [T_v]_\beta^{(1_{\mathbb{F}})}\begin{bmatrix} 1 \end{bmatrix}.$$

Since multiplying an $n \times 1$ matrix by $\begin{bmatrix} 1 \end{bmatrix}$ returns its unique column, that unique column must be $[v]_\beta$. Hence $[T_v]_\beta^{(1_{\mathbb{F}})} = [v]_\beta$.

$$\boxed{[T_v]_\beta^{(1_{\mathbb{F}})} = [v]_\beta}.$$

Question 2 (Nested span condition $\Rightarrow$ upper-triangular matrix)

Problem

Let

(i) V and W be finite-dimensional vector spaces over a field $\mathbb{F}$, with $\dim(V) = \dim(W) = n$,

(ii) $\beta = (v_1, \ldots, v_n)$ and $\gamma = (w_1, \ldots, w_n)$ be ordered bases for V and W, respectively, and

(iii) $T \in \mathcal{L}(V, W)$ with the property that

$$\forall k \in \{1, \ldots, n\} : \qquad T(v_k) \in \operatorname{span}(\{w_1, \ldots, w_k\}).$$

Show that $[T]_\gamma^\beta$ is an upper-triangular $n \times n$ $\mathbb{F}$-matrix.

Solution

Method 1: Column-by-column argument (coordinates of $T(v_k)$)

Let $A = [T]_\gamma^\beta = (a_{ij})_{1 \le i,j \le n}$. By definition, the kth column of A is

$$[T(v_k)]_\gamma = \begin{bmatrix} a_{1k} \\ \vdots \\ a_{nk} \end{bmatrix}.$$

The hypothesis $T(v_k) \in \operatorname{span}(w_1, \ldots, w_k)$ means precisely that in the γ-coordinates,

$$T(v_k) = \sum_{i=1}^{k} a_{ik} w_i,$$

so the coefficients in front of $w_{k+1}, \ldots, w_n$ must be 0. Equivalently,

$$a_{ik} = 0 \quad \text{for all } i > k.$$

But $a_{ik} = 0$ for $i > k$ says: all entries strictly below the diagonal are 0. Therefore A is upper triangular.

$$\boxed{[T]_\gamma^\beta \text{ is upper triangular.}}$$

Method 2: Coordinate functionals (dual basis)

Let $A = [T]_\gamma^\beta = (a_{ij})_{1\le i,j\le n}$. Fix $k \in \{1, \dots, n\}$. Since $\gamma = (w_1, \dots, w_n)$ is a basis, there exist $\varphi_i \in \mathcal{L}(W, \mathbb{F})$ such that

$$\varphi_i(w_j) = \delta_{ij}.$$

Then

$$a_{ik} = \varphi_i\big(T(v_k)\big).$$

Given $T(v_k) \in \operatorname{span}(w_1, \dots, w_k)$, we have

$$\varphi_i\big(T(v_k)\big) = 0 \quad \text{for all } i > k.$$

Hence $a_{ik} = 0$ for all $i > k$, so $[T]_\gamma^\beta$ is upper triangular.

$$\boxed{[T]_\gamma^\beta \text{ is upper triangular.}}$$

Method 3: Flag invariance

Let $A = [T]_\gamma^\beta = (a_{ij})_{1\le i,j\le n}$, $\quad W_k := \operatorname{span}(w_1, \dots, w_k)$. Then

$$T(v_k) \in W_k \implies T\big(\operatorname{span}(v_1, \dots, v_k)\big) \subseteq W_k.$$

In basis γ, vectors in W_k have zero coordinates below row k. Hence the kth column of $[T]_\gamma^\beta$ has zeros below row k, i.e.

$$a_{ik} = 0 \quad (i > k).$$

Thus $[T]_\gamma^\beta$ is upper triangular.

$$\boxed{[T]_\gamma^\beta \text{ is upper triangular.}}$$

Method 4: Filtration / quotient argument

Let $V_k := \operatorname{span}(v_1, \dots, v_k)$ and $W_k := \operatorname{span}(w_1, \dots, w_k)$. Then

$$T(V_k) \subseteq W_k \quad \forall k.$$

This induces linear maps

$$\widetilde{T}_k : V_k/V_{k-1} \to W_k/W_{k-1},$$

each between 1-dimensional spaces. Hence each $\widetilde{T}_k$ is scalar, implying no mixing into lower levels.

Thus $a_{ik} = 0$ for all $i > k$, so $[T]_\gamma^\beta$ is upper triangular.

$$\boxed{[T]_\gamma^\beta \text{ is upper triangular.}}$$

Question 3 (Right-shift on $\mathbb{R}^\infty$)

Problem

Recall the right-shift operator $\sigma_\rightarrow \in \mathcal{L}(\mathbb{R}^\infty)$, defined by

$$\forall x = (x_1, x_2, x_3, \dots) \in \mathbb{R}^\infty : \qquad \sigma_\rightarrow(x) \coloneqq (0, x_1, x_2, x_3, \dots).$$

(a) Show that $\sigma_\rightarrow$ is injective but not surjective onto $\mathbb{R}^\infty$.

(b) Why does this not contradict Problem 3 of Tutorial Problem Set 6?

(c) Show that $\sigma_\rightarrow$ has no (real) eigenvalues.

Solution

Method 1: Direct definition-chasing on sequences

(a) **Injective.** Suppose $\sigma_\rightarrow(x) = \sigma_\rightarrow(y)$ for $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$. Then

$$(0, x_1, x_2, x_3, \dots) = (0, y_1, y_2, y_3, \dots),$$

so $x_k = y_k$ for all $k \geq 1$. Hence $x = y$, proving injectivity.

Not surjective. Consider $z = (1, 0, 0, 0, \dots) \in \mathbb{R}^\infty$. If z were in the range of $\sigma_\rightarrow$, then $z = \sigma_\rightarrow(x) = (0, x_1, x_2, \dots)$ for some x, which forces $z_1 = 0$, contradiction since $z_1 = 1$. Hence $\sigma_\rightarrow$ is not surjective.

$$\boxed{\sigma_\rightarrow \text{ injective but not surjective on } \mathbb{R}^\infty.}$$

(b) Problem 3 of Tutorial Problem Set 6 established (in finite dimensions) that if $\dim(V) = \dim(W) < \infty$, then injective $\Leftrightarrow$ surjective for $T \in \mathcal{L}(V, W)$. Here $\mathbb{R}^\infty$ is *infinite-dimensional*. The finite-dimensional argument relies on rank–nullity / dimension counting, which does not apply as stated when dimensions are infinite. Therefore there is no contradiction.

$$\boxed{\text{No contradiction: } \mathbb{R}^\infty \text{ is infinite-dimensional, so injective} \not\Rightarrow \text{surjective.}}$$

(c) Suppose $\lambda \in \mathbb{R}$ is an eigenvalue. Then there exists $x \neq 0$ in $\mathbb{R}^\infty$ such that

$$\sigma_\rightarrow(x) = \lambda x.$$

Writing coordinates:

$$(0, x_1, x_2, x_3, \dots) = (\lambda x_1, \lambda x_2, \lambda x_3, \dots).$$

Comparing the first coordinate gives $0 = \lambda x_1$.

If $\lambda \neq 0$, then $x_1 = 0$. Comparing the second coordinate gives $x_1 = \lambda x_2$, hence $0 = \lambda x_2$ and so $x_2 = 0$. Inductively, $x_k = 0$ for all k, so $x = 0$, contradicting the eigenvector requirement $x \neq 0$.

If $\lambda = 0$, then $\sigma_{\rightarrow}(x) = 0$ implies $(0, x_1, x_2, \dots) = (0, 0, 0, \dots)$, hence $x = 0$, again a contradiction.

Thus no $\lambda \in \mathbb{R}$ can satisfy the eigenvalue condition.

$$\boxed{\sigma_{\rightarrow} \text{ has no real eigenvalues.}}$$

Method 2: Kernel/range structure and eigenvalue obstruction

(a) $\ker(\sigma_{\rightarrow}) = \{0\}$ because $\sigma_{\rightarrow}(x) = 0$ forces all $x_k = 0$. Hence $\sigma_{\rightarrow}$ is injective. But $\operatorname{range}(\sigma_{\rightarrow}) = \{(y_1, y_2, \dots) : y_1 = 0\}$, a proper subset of $\mathbb{R}^\infty$, so it is not surjective.

(b) The injective $\Rightarrow$ surjective equivalence holds for finite-dimensional spaces of the same dimension; $\mathbb{R}^\infty$ is not finite-dimensional.

(c) If $\lambda = 0$ were an eigenvalue, we would need $\ker(\sigma_{\rightarrow}) \neq \{0\}$, but $\ker(\sigma_{\rightarrow}) = \{0\}$. If $\lambda \neq 0$ were an eigenvalue with eigenvector x, then $x = \lambda^{-1}\sigma_{\rightarrow}(x)$ lies in $\operatorname{range}(\sigma_{\rightarrow})$, so $x_1 = 0$. Applying the same reasoning repeatedly gives $x \in \operatorname{range}(\sigma_{\rightarrow}^k)$ for all k, forcing $x_1 = x_2 = \cdots = 0$ and thus $x = 0$, contradiction. Hence no real eigenvalues exist.

Question 4 (Rotation on $\mathbb{R}^2$ has no real eigenvalues unless $\theta \in \mathbb{Z}\pi$)

Problem

Let $\theta \in \mathbb{R}$, and define $R_\theta \in \mathcal{L}(\mathbb{R}^2)$ by

$$\forall x, y \in \mathbb{R} : \qquad R_\theta(x, y) := (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta).$$

Show that if $\theta \in \mathbb{R} \setminus \mathbb{Z}\pi$, then R_θ has no (real) eigenvalues.

Remark: R_θ is the operator of counterclockwise rotation by the angle θ.

Solution

Method 1: Characteristic polynomial and discriminant

With respect to the standard basis, R_θ has matrix

$$A_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

A real number λ is an eigenvalue iff $\det(A_\theta - \lambda I_2) = 0$:

$$\begin{aligned} 0 = \det \begin{bmatrix} \cos\theta - \lambda & -\sin\theta \\ \sin\theta & \cos\theta - \lambda \end{bmatrix} &= (\cos\theta - \lambda)^2 + \sin^2\theta \\ &= \lambda^2 - 2(\cos\theta)\lambda + (\cos^2\theta + \sin^2\theta) \\ &= \lambda^2 - 2(\cos\theta)\lambda + 1. \end{aligned}$$

Thus eigenvalues (over $\mathbb{C}$) are

$$\lambda = \cos\theta \pm \sqrt{\cos^2\theta - 1} = \cos\theta \pm i\sin\theta.$$

These are real iff $\sin\theta = 0$, i.e. $\theta \in \mathbb{Z}\pi$. Hence if $\theta \notin \mathbb{Z}\pi$, then $\sin\theta \neq 0$ and there are no real eigenvalues.

$$\boxed{\theta \notin \mathbb{Z}\pi \implies R_\theta \text{ has no real eigenvalues.}}$$

Method 2: Eigenvector implies invariant line

Assume $\lambda \in \mathbb{R}$ is an eigenvalue, with eigenvector $u \in \mathbb{R}^2 \setminus \{0\}$:

$$R_\theta(u) = \lambda u.$$

Then $R_\theta(u) \in \operatorname{span}(u)$, so $R_\theta(u)$ is parallel to u.

By definition, R_θ rotates every nonzero vector through angle θ. A nonzero vector is sent to a parallel vector by rotation iff the rotation angle is 0 or π modulo 2π, i.e.

$$\theta \in \mathbb{Z}\pi.$$

Hence $\theta \notin \mathbb{Z}\pi$ contradicts the existence of such u.

Therefore R_θ has no real eigenvalues.

$$\boxed{\theta \notin \mathbb{Z}\pi \implies R_\theta \text{ has no real eigenvalues.}}$$

Method 3: Norm preservation

Assume $R_\theta(u) = \lambda u$ for some $u \neq 0$ and $\lambda \in \mathbb{R}$.

Since R_θ is a rotation,

$$\|R_\theta(u)\| = \|u\|.$$

But

$$\|R_\theta(u)\| = \|\lambda u\| = |\lambda|\,\|u\|.$$

As $u \neq 0$, this gives

$$|\lambda| = 1 \implies \lambda = \pm 1.$$

Hence

$$R_\theta(u) = \pm u,$$

so $R_\theta(u)$ is parallel to u. A rotation sends a nonzero vector to a parallel vector iff $\theta \in \mathbb{Z}\pi$. Contradiction.

Thus $\theta \notin \mathbb{Z}\pi \Rightarrow R_\theta$ has no real eigenvalues.

$$\boxed{\theta \notin \mathbb{Z}\pi \implies R_\theta \text{ has no real eigenvalues.}}$$

Method 4: Trace–determinant criterion

With respect to the standard basis,

$$[R_\theta] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \operatorname{tr}(R_\theta) = 2\cos\theta, \qquad \det(R_\theta) = 1.$$

Hence any eigenvalue λ satisfies

$$\lambda^2 - \operatorname{tr}(R_\theta)\lambda + \det(R_\theta) = 0,$$

i.e.

$$\lambda^2 - 2(\cos\theta)\lambda + 1 = 0.$$

Its discriminant is

$$\Delta = 4\cos^2\theta - 4 = -4\sin^2\theta.$$

If $\theta \notin \mathbb{Z}\pi$, then $\sin\theta \neq 0$, so

$$\Delta < 0.$$

Therefore the quadratic has no real root, so R_θ has no real eigenvalues.

$$\boxed{\theta \notin \mathbb{Z}\pi \implies R_\theta \text{ has no real eigenvalues.}}$$

Method 5: Complexification

Identify $\mathbb{R}^2$ with $\mathbb{C}$ via

$$(x, y) \longleftrightarrow x + iy.$$

Under this identification,

$$R_\theta(z) = e^{i\theta}z \qquad (z \in \mathbb{C}).$$

Thus, over $\mathbb{C}$, the scalar corresponding to R_θ is

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

A real eigenvalue would have to be real as a complex number. But

$$e^{i\theta} \in \mathbb{R} \iff \sin\theta = 0 \iff \theta \in \mathbb{Z}\pi.$$

Hence if $\theta \notin \mathbb{Z}\pi$, then $e^{i\theta} \notin \mathbb{R}$, so R_θ has no real eigenvalues.

$$\boxed{\theta \notin \mathbb{Z}\pi \implies R_\theta \text{ has no real eigenvalues.}}$$

Question 5 (Eigenvalues/eigenspaces of a map on $\mathbb{R}^{2,1}$)

Problem

Define $F \in \mathcal{L}(\mathbb{R}^{2,1})$ by

$$\forall x, y \in \mathbb{R} : \qquad F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) := \begin{bmatrix} y \\ x + y \end{bmatrix}.$$

(a) Compute the (real) eigenvalues of F.

(b) For each (real) eigenvalue λ of F, determine its corresponding eigenspace $E_{F,\lambda}$.

Solution

Method 1: Matrix/characteristic polynomial

With respect to the standard basis of $\mathbb{R}^{2,1}$,

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}, \qquad A := \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

(a) The characteristic polynomial is

$$\chi_A(\lambda) = \det(A - \lambda I_2) = \det \begin{bmatrix} -\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix} = \lambda^2 - \lambda - 1.$$

Thus the (real) eigenvalues are the roots:

$$\lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2}.$$

$$\boxed{\text{Eigenvalues:} \quad \lambda = \frac{1+\sqrt{5}}{2}, \ \frac{1-\sqrt{5}}{2}}.$$

(b) For an eigenvalue λ, solve $(A - \lambda I_2)\binom{x}{y} = 0$:

$$\begin{bmatrix} -\lambda & 1 \\ 1 & 1-\lambda \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \Longrightarrow \quad -\lambda x + y = 0 \ \Rightarrow \ y = \lambda x.$$

(Then the second equation is $x + (1-\lambda)y = 0$, which holds automatically because $\lambda^2 - \lambda - 1 = 0$.)

Hence an eigenvector is $\binom{1}{\lambda}$ and the eigenspace is one-dimensional:

$$E_{F,\lambda} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right\}.$$

So

$$\boxed{E_{F,\lambda} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right\} \text{ for } \lambda \in \left\{\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}\right\}}.$$

Method 2: Solve the eigenvector equation directly

Let $\lambda \in \mathbb{R}$ and $u = \binom{x}{y} \neq 0$ satisfy

$$F(u) = \lambda u.$$

Then

$$\begin{bmatrix} y \\ x+y \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix},$$

so

$$y = \lambda x, \qquad x + y = \lambda y.$$

Substitute $y = \lambda x$ into the second equation:

$$x + \lambda x = \lambda^2 x.$$

If $x = 0$, then $y = \lambda x = 0$, contradicting $u \neq 0$. Hence $x \neq 0$, and dividing by x gives

$$\lambda^2 - \lambda - 1 = 0.$$

Thus

$$\lambda = \frac{1 \pm \sqrt{5}}{2}.$$

For either such λ, the relation $y = \lambda x$ gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ \lambda \end{bmatrix}.$$

Therefore

$$E_{F,\lambda} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ \lambda \end{bmatrix} \right\}, \qquad \lambda \in \left\{ \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2} \right\}.$$

$$\boxed{\text{Eigenvalues: } \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}}$$

$$\boxed{E_{F,\lambda} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ \lambda \end{bmatrix} \right\} \text{ for each such } \lambda.}$$

Method 3: Trace–determinant shortcut

With respect to the standard basis,

$$[F] = A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \operatorname{tr}(A) = 1, \qquad \det(A) = -1.$$

Hence any eigenvalue λ satisfies

$$\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0,$$

i.e.

$$\lambda^2 - \lambda - 1 = 0.$$

So

$$\lambda = \frac{1 \pm \sqrt{5}}{2}.$$

Now let λ be either root. An eigenvector $u = \binom{x}{y} \neq 0$ satisfies

$$F(u) = \lambda u \implies y = \lambda x.$$

If $x = 0$, then $y = 0$, impossible. Thus $x \neq 0$, and

$$u = x \begin{bmatrix} 1 \\ \lambda \end{bmatrix}.$$

Therefore

$$E_{F,\lambda} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ \lambda \end{bmatrix} \right\}.$$

$$\boxed{\text{Eigenvalues: } \frac{1 + \sqrt{5}}{2}, \ \frac{1 - \sqrt{5}}{2}}$$

$$\boxed{E_{F,\lambda} = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ \lambda \end{bmatrix} \right\} \text{ for each such } \lambda.}$$

Method 4: Ansatz $\binom{1}{r}$

Seek an eigenvector of the form

$$\begin{bmatrix} 1 \\ r \end{bmatrix}.$$

Then

$$F\left(\begin{bmatrix} 1 \\ r \end{bmatrix}\right) = \begin{bmatrix} r \\ 1+r \end{bmatrix}.$$

If this is an eigenvector with eigenvalue λ, then

$$\begin{bmatrix} r \\ 1+r \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ r \end{bmatrix},$$

hence

$$r = \lambda, \qquad 1 + r = \lambda r.$$

Substituting $r = \lambda$ gives

$$1 + \lambda = \lambda^2 \quad \Longleftrightarrow \quad \lambda^2 - \lambda - 1 = 0.$$

Thus

$$\lambda = \frac{1 \pm \sqrt{5}}{2}.$$

For either such λ, taking $r = \lambda$ shows directly that

$$F\left(\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right) = \lambda \begin{bmatrix} 1 \\ \lambda \end{bmatrix},$$

so

$$E_{F,\lambda} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right\}.$$

$$\boxed{\text{Eigenvalues:} \quad \frac{1+\sqrt{5}}{2}, \ \frac{1-\sqrt{5}}{2}}$$

$$\boxed{E_{F,\lambda} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right\} \ \text{for each such } \lambda.}$$

Method 5: Recognise the Fibonacci / companion pattern

With respect to the standard basis,

$$[F] = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

This is the standard Fibonacci matrix, equivalently the companion matrix of

$$t^2 - t - 1.$$

Hence its characteristic polynomial is

$$\chi_F(\lambda) = \lambda^2 - \lambda - 1,$$

so the real eigenvalues are

$$\lambda = \frac{1 \pm \sqrt{5}}{2}.$$

For an eigenvalue λ, the eigenvector equation

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$$

gives

$$y = \lambda x.$$

Thus every eigenvector has the form

$$\begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ \lambda \end{bmatrix},$$

and hence

$$E_{F,\lambda} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right\}.$$

$$\boxed{\text{Eigenvalues: } \frac{1+\sqrt{5}}{2},\ \frac{1-\sqrt{5}}{2}}$$

$$\boxed{E_{F,\lambda} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ \lambda \end{bmatrix}\right\} \text{ for each such } \lambda.}$$

Question 6 (Eigen-structure of $T(A) = A^\mathsf{T} + 2\,\mathrm{tr}(A)I_2$ on $\mathbb{R}^{2,2}$)

Problem

Define $T \in \mathcal{L}(\mathbb{R}^{2,2})$ by

$$\forall A \in \mathbb{R}^{2,2}: \qquad T(A) := A^\mathsf{T} + 2\ \mathrm{tr}(A)\, I_2.$$

(a) Compute the (real) eigenvalues of T.

(b) For each (real) eigenvalue λ of T, determine its corresponding eigenspace $E_{T,\lambda}$.

(c) Does $\mathbb{R}^{2,2}$ have a basis consisting of eigenvectors of T? If so, derive such a basis.

Solution

Method 1: Write T in coordinates and compute eigenvalues/eigenspaces

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \mathrm{tr}(A) = a + d, \qquad A^\mathsf{T} = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

Then

$$T(A) = A^\mathsf{T} + 2(a+d)I_2 = \begin{bmatrix} a + 2(a+d) & c \\ b & d + 2(a+d) \end{bmatrix} = \begin{bmatrix} 3a + 2d & c \\ b & 2a + 3d \end{bmatrix}.$$

Thus in the coordinate vector $(a, b, c, d)^\mathsf{T}$ (relative to the standard basis $(E_{11}, E_{12}, E_{21}, E_{22})$), the operator has matrix

$$M = \begin{bmatrix} 3 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 3 \end{bmatrix}, \quad \text{since } (a, b, c, d) \mapsto (3a + 2d,\ c,\ b,\ 2a + 3d).$$

(a) The matrix M is block diagonal with respect to the decomposition

$$(a, d) \text{ block: } \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}, \qquad (b, c) \text{ block: } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Eigenvalues of $\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ are 5 and 1; eigenvalues of $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ are 1 and -1. Hence the real eigenvalues of T are

$$\boxed{\lambda \in \{-1,\ 1,\ 5\}}$$

(with $\lambda = 1$ having algebraic multiplicity 2).

(b) Compute eigenspaces by solving $T(A) = \lambda A$.

Case $\lambda = 5$. From $T(A) = 5A$, comparing off-diagonals gives $c = 5b$ and $b = 5c$, hence $b = c = 0$. Comparing diagonals gives

$$3a + 2d = 5a \Rightarrow d = a, \qquad 2a + 3d = 5d \Rightarrow a = d,$$

consistent. Thus $A = aI_2$. Hence

$$\boxed{E_{T,5} = \operatorname{span}\{I_2\}}.$$

Case $\lambda = -1$. Then $T(A) = -A$ gives diagonals $3a + 2d = -a$ and $2a + 3d = -d$, i.e. $4a + 2d = 0$ and $2a + 4d = 0$, so $a = d = 0$. Off-diagonals give $c = -b$ and $b = -c$ (same condition). Thus

$$A = b\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

So

$$\boxed{E_{T,-1} = \operatorname{span}\left\{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right\}}.$$

Case $\lambda = 1$. Then $T(A) = A$ gives diagonals $3a + 2d = a$ and $2a + 3d = d$, i.e. $2a + 2d = 0$ so $d = -a$. Off-diagonals give $c = b$ and $b = c$, i.e. $b = c$ is free. Hence

$$A = a\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Therefore

$$\boxed{E_{T,1} = \operatorname{span}\left\{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right\} \text{ (dimension 2).}}$$

(c) Since

$$\dim(E_{T,5}) + \dim(E_{T,1}) + \dim(E_{T,-1}) = 1 + 2 + 1 = 4 = \dim(\mathbb{R}^{2,2}),$$

the eigenvectors span $\mathbb{R}^{2,2}$, so $\mathbb{R}^{2,2}$ has a basis consisting of eigenvectors of T. One such eigenbasis is

$$\left(I_2, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right),$$

corresponding to eigenvalues $5, 1, 1, -1$, respectively.

$$\boxed{\text{Yes. An eigenbasis is } \left\{I_2, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\right\}.}$$

Method 2: Decompose into symmetric and skew-symmetric parts

Let
$$\mathrm{Sym}_2 := \{A \in \mathbb{R}^{2,2} : A^\mathsf{T} = A\}, \qquad \mathrm{Skew}_2 := \{A \in \mathbb{R}^{2,2} : A^\mathsf{T} = -A\}.$$
Then
$$\mathbb{R}^{2,2} = \mathrm{Sym}_2 \oplus \mathrm{Skew}_2 .$$

If $A \in \mathrm{Skew}_2$, then $\mathrm{tr}(A) = 0$ and $A^\mathsf{T} = -A$, so
$$T(A) = A^\mathsf{T} + 2\,\mathrm{tr}(A)I_2 = -A.$$
Hence
$$E_{T,-1} = \mathrm{Skew}_2 = \mathrm{span}\left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}.$$

If $A \in \mathrm{Sym}_2$, then $A^\mathsf{T} = A$, so
$$T(A) = A + 2\,\mathrm{tr}(A)I_2.$$
Now split
$$\mathrm{Sym}_2 = \mathbb{R}I_2 \oplus \mathrm{Sym}_2^0, \qquad \mathrm{Sym}_2^0 := \{A \in \mathrm{Sym}_2 : \mathrm{tr}(A) = 0\}.$$
Then:
$$A = tI_2 \implies T(A) = tI_2 + 2(2t)I_2 = 5tI_2,$$
so
$$E_{T,5} = \mathbb{R}I_2 = \mathrm{span}\{I_2\},$$
and
$$A \in \mathrm{Sym}_2^0 \implies T(A) = A,$$
so
$$E_{T,1} = \mathrm{Sym}_2^0 = \mathrm{span}\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}.$$

Thus
$$E_{T,5} = \mathbb{R}I_2, \qquad E_{T,1} = \mathrm{Sym}_2^0, \qquad E_{T,-1} = \mathrm{Skew}_2 .$$
Their dimensions are $1, 2, 1$, summing to 4, so T is diagonalizable. Hence
$$\left(I_2, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right)$$
is an eigenbasis.

$$\boxed{\lambda = -1, 1, 5;\ E_{T,-1} = \mathrm{Skew}_2,\ E_{T,1} = \mathrm{Sym}_2^0,\ E_{T,5} = \mathbb{R}I_2.}$$

Method 3: Canonical decomposition $\mathbb{R}^{2,2} = \mathbb{R}I_2 \oplus \mathrm{Sym}_2^0 \oplus \mathrm{Skew}_2$

Use the direct sum decomposition

$$\mathbb{R}^{2,2} = \mathbb{R}I_2 \oplus \mathrm{Sym}_2^0 \oplus \mathrm{Skew}_2,$$

where

$$\mathrm{Sym}_2^0 := \{A \in \mathbb{R}^{2,2} : A^\mathsf{T} = A,\ \mathrm{tr}(A) = 0\}, \qquad \mathrm{Skew}_2 := \{A \in \mathbb{R}^{2,2} : A^\mathsf{T} = -A\}.$$

Let

$$A = tI_2 + S + K \qquad (t \in \mathbb{R},\ S \in \mathrm{Sym}_2^0,\ K \in \mathrm{Skew}_2).$$

Then

$$A^\mathsf{T} = tI_2 + S - K, \qquad \mathrm{tr}(A) = 2t.$$

Hence

$$T(A) = A^\mathsf{T} + 2\,\mathrm{tr}(A)I_2 = (tI_2 + S - K) + 4tI_2 = 5tI_2 + S - K.$$

Therefore T acts by scalar multiplication on the three summands:

$$5 \text{ on } \mathbb{R}I_2, \qquad 1 \text{ on } \mathrm{Sym}_2^0, \qquad -1 \text{ on } \mathrm{Skew}_2\,.$$

So the real eigenvalues are

$$\boxed{-1,\ 1,\ 5}.$$

Moreover,

$$E_{T,5} = \mathbb{R}I_2, \qquad E_{T,1} = \mathrm{Sym}_2^0, \qquad E_{T,-1} = \mathrm{Skew}_2\,.$$

A concrete basis for these eigenspaces is

$$E_{T,5} = \mathrm{span}\{I_2\},$$

$$E_{T,1} = \mathrm{span}\left\{\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\right\},$$

$$E_{T,-1} = \mathrm{span}\left\{\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}\right\}.$$

Thus T is diagonalizable, and an eigenbasis is

$$\left(I_2, \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}\right).$$

$$\boxed{T \text{ is diagonalizable with eigenspaces } \mathbb{R}I_2,\ \mathrm{Sym}_2^0,\ \mathrm{Skew}_2\,.}$$

Method 4: Choose a basis adapted to the operator

Consider

$$B_1 := I_2, \qquad B_2 := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad B_3 := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad B_4 := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

These four matrices are linearly independent, hence form a basis of $\mathbb{R}^{2,2}$.

Now compute:

$$T(B_1) = B_1 + 2\,\mathrm{tr}(B_1) I_2 = I_2 + 4I_2 = 5B_1,$$

since $\mathrm{tr}(B_1) = 2$.

Also B_2, B_3 are symmetric and traceless, so

$$T(B_2) = B_2, \qquad T(B_3) = B_3.$$

Finally, B_4 is skew-symmetric and traceless, so

$$T(B_4) = -B_4.$$

Hence B_1, B_2, B_3, B_4 are eigenvectors with eigenvalues

$$5,\ 1,\ 1,\ -1,$$

respectively. Therefore the real eigenvalues are

$$\boxed{-1,\ 1,\ 5}.$$

Their eigenspaces are

$$E_{T,5} = \mathrm{span}\{B_1\} = \mathrm{span}\{I_2\},$$

$$E_{T,1} = \mathrm{span}\{B_2, B_3\} = \mathrm{span}\left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\},$$

$$E_{T,-1} = \mathrm{span}\{B_4\} = \mathrm{span}\left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}.$$

Since (B_1, B_2, B_3, B_4) is a basis of eigenvectors, T is diagonalizable.

$$\boxed{\text{Yes. } (B_1, B_2, B_3, B_4) \text{ is an eigenbasis of } \mathbb{R}^{2,2}.}$$

Method 5: Projection-operator viewpoint

Every $A \in \mathbb{R}^{2,2}$ decomposes uniquely as

$$A = \alpha I_2 + S + K,$$

where

$$\alpha := \frac{\operatorname{tr}(A)}{2}, \qquad S := \frac{A + A^\mathsf{T}}{2} - \frac{\operatorname{tr}(A)}{2} I_2 \in \operatorname{Sym}_2^0, \qquad K := \frac{A - A^\mathsf{T}}{2} \in \operatorname{Skew}_2 .$$

Then

$$A^\mathsf{T} = \alpha I_2 + S - K, \qquad \operatorname{tr}(A) = 2\alpha.$$

So

$$T(A) = A^\mathsf{T} + 2\operatorname{tr}(A) I_2 = (\alpha I_2 + S - K) + 4\alpha I_2 = 5\alpha I_2 + S - K.$$

Thus the three components are eigen-components:

$$\alpha I_2 \mapsto 5\alpha I_2, \qquad S \mapsto S, \qquad K \mapsto -K.$$

Hence

$$E_{T,5} = \mathbb{R} I_2, \qquad E_{T,1} = \operatorname{Sym}_2^0, \qquad E_{T,-1} = \operatorname{Skew}_2 .$$

In particular, the real eigenvalues are

$$\boxed{-1,\ 1,\ 5}.$$

Choosing bases of these three subspaces gives an eigenbasis; for example,

$$\left(I_2, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right).$$

$$\boxed{T \text{ acts diagonally on } \mathbb{R} I_2 \oplus \operatorname{Sym}_2^0 \oplus \operatorname{Skew}_2, \text{ so } T \text{ is diagonalizable.}}$$

Question 7 (Eigenvalues/eigenspaces of an invertible operator and its inverse)

Problem

Let

(i) V be a vector space over a field $\mathbb{F}$ (not necessarily finite-dimensional), and

(ii) $T \in \mathcal{L}(V)$ be invertible.

Establish the following two statements:

(1) If $\lambda \in \mathbb{F}$ is an eigenvalue of T, then $\lambda \neq 0_{\mathbb{F}}$ and λ^{-1} is an eigenvalue of T^{-1}.

(2) If $\lambda \in \mathbb{F}$ is an eigenvalue of T, then

$$E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}.$$

Solution

Method 1: Apply T^{-1} to the eigenvector

Let $\lambda \in \mathbb{F}$ be an eigenvalue of T. Then there exists $v \in V \setminus \{0\}$ such that

$$T(v) = \lambda v.$$

(1) Since T is invertible, $\ker(T) = \{0\}$. If $\lambda = 0$, then

$$T(v) = \lambda v = 0,$$

which would imply $v \in \ker(T)$, hence $v = 0$, contradiction. Therefore

$$\lambda \neq 0_{\mathbb{F}}.$$

Now apply T^{-1} to $T(v) = \lambda v$:

$$v = T^{-1}(\lambda v) = \lambda\, T^{-1}(v).$$

Since $\lambda \neq 0$, multiply by λ^{-1}:

$$T^{-1}(v) = \lambda^{-1} v.$$

Thus $v \neq 0$ is an eigenvector of T^{-1} with eigenvalue λ^{-1}. Hence

$$\boxed{\lambda^{-1} \text{ is an eigenvalue of } T^{-1}.}$$

(2) Let $v \in E_{T,\lambda}$. Then $v \neq 0$ and $T(v) = \lambda v$. By part (1),

$$T^{-1}(v) = \lambda^{-1} v,$$

so $v \in E_{T^{-1},\lambda^{-1}}$. Thus

$$E_{T,\lambda} \subseteq E_{T^{-1},\lambda^{-1}}.$$

Conversely, let $v \in E_{T^{-1},\lambda^{-1}}$. Then $v \neq 0$ and

$$T^{-1}(v) = \lambda^{-1}v.$$

Apply T to both sides:

$$v = \lambda^{-1}T(v).$$

Since $\lambda \neq 0$, this gives

$$T(v) = \lambda v.$$

Hence $v \in E_{T,\lambda}$, so

$$E_{T^{-1},\lambda^{-1}} \subseteq E_{T,\lambda}.$$

Therefore

$$\boxed{E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}.}$$

Method 2: Rearrangement of the eigenvalue equation

Let $v \neq 0$ satisfy

$$T(v) = \lambda v.$$

(1) If $\lambda = 0$, then $T(v) = 0$, contradicting invertibility of T. Hence $\lambda \neq 0$.

Also,

$$T(v) = \lambda v \quad \iff \quad v = \lambda\, T^{-1}(v) \quad \iff \quad T^{-1}(v) = \lambda^{-1}v.$$

Thus λ^{-1} is an eigenvalue of T^{-1}.

(2) For any $v \neq 0$,

$$T(v) = \lambda v \quad \iff \quad T^{-1}(v) = \lambda^{-1}v.$$

Hence

$$v \in E_{T,\lambda} \quad \iff \quad v \in E_{T^{-1},\lambda^{-1}}.$$

Therefore

$$\boxed{E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}.}$$

Method 3: Kernel formulation

Recall
$$E_{T,\lambda} = \ker(T - \lambda I) \setminus \{0\}, \qquad E_{T^{-1},\lambda^{-1}} = \ker(T^{-1} - \lambda^{-1} I) \setminus \{0\}.$$

Let λ be an eigenvalue of T. Then $\ker(T - \lambda I) \neq \{0\}$.

(1) Since T is invertible, 0 cannot be an eigenvalue of T, so $\lambda \neq 0$.

Now
$$T^{-1} - \lambda^{-1} I = -\lambda^{-1} T^{-1}(T - \lambda I).$$

Therefore
$$\ker(T^{-1} - \lambda^{-1} I) = \ker(T - \lambda I),$$

because $-\lambda^{-1}T^{-1}$ is invertible. Hence $\ker(T^{-1} - \lambda^{-1} I) \neq \{0\}$, so λ^{-1} is an eigenvalue of T^{-1}.

(2) From
$$\ker(T^{-1} - \lambda^{-1} I) = \ker(T - \lambda I),$$

we immediately obtain
$$E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}.$$

$$\boxed{\lambda \neq 0,\ \lambda^{-1} \in \sigma(T^{-1}),\ \text{and } E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}.}$$

Method 4: Operator identity

Let λ be an eigenvalue of T. Since T is invertible, $\lambda \neq 0$.

Observe the identity
$$T - \lambda I = -\lambda\, T\big(T^{-1} - \lambda^{-1} I\big).$$

Since $-\lambda T$ is invertible, the two operators have the same kernel:
$$\ker(T - \lambda I) = \ker(T^{-1} - \lambda^{-1} I).$$

Hence
$$E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}.$$

In particular, if $E_{T,\lambda} \neq \{0\}$, then also
$$E_{T^{-1},\lambda^{-1}} \neq \{0\},$$

so λ^{-1} is an eigenvalue of T^{-1}.

$$\boxed{E_{T,\lambda} = E_{T^{-1},\lambda^{-1}}, \quad \lambda^{-1} \text{ is an eigenvalue of } T^{-1}.}$$